



บริษัท โฟมไทย คอร์ปอเรชั่น จำกัด
FOMOTHAI CORPORATION COMPANY LIMITED

MATERIAL SAFETY DATA SHEET OF
CLOMAZONE + PROPANIL 12% + 27% W/V EC

SECTION 1: CHEMICAL PRODUCT/COMPANY IDENTIFICATION

FOMOTHAI CORPORATION CO., LTD.

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Product name : Clomazone + Propanil 12% + 27% W/V EC

Common name : clomazone
propanil

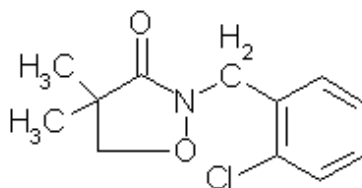
Chemical Name : (clomazone) : 2-(2-chlorobenzyl)-4,4-dimethyl-1,2-oxazolidin-3-one;
2-(2-chlorobenzyl)-4,4-dimethylisoxazolidin-3-one
(propanil) : N-(3,4-Dichlorophenyl) propanamide

CAS Number : (clomazone) : 8177-89-1
(propanil) : 709-98-8

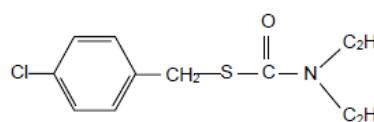
Molecular formula : (clomazone) : $C_{12}H_{14}ClNO_2$
(propanil) : $C_9H_9OCl_2N$

Molecular Weight : (clomazone) : 239.72
(propanil) : 218.08

Structural formula :



(clomazone)



(propanil)

Type of formulation : Emulsifiable concentrate (EC)

Use category : Herbicide

SECTION 2: COMPOSITION / INFORMATION ON INGREDIENTS

Composition	CAS no.	Content %
Clomazone	8177-89-1	12
Propanil	709-98-8	27
Others ingredients	- -	Balance



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SECTION 3: HAZARDS IDENTIFICATION

Harmful by inhalation. Irritating to eyes, respiratory system and skin. Risk of serious damage to eyes

SECTION 4: FIRST AID MEASURES

- | | | |
|--------------|---|---|
| Ingestion | : | Induce vomiting after drinking water. As soon as vomited fluid is clear, administer as much milk as the patient can tolerate. Avoid fats, oils, alcohol or epinephrine. Get prompt medical attention. |
| Eye contact | : | Flush with plenty of clear water for 15-20 minutes. |
| Skin contact | : | Remove contaminated clothing and wash skin thoroughly with soap and water |
| Inhalation | : | Move affect person to fresh air and keep at rest until recovered. If not breathing, give artificial respiration and get to a doctor. |

SECTION 5: FIRE FIGHTING MEASURES

- | | | |
|---------------------------|---|---|
| Extinguishing Media | : | Water spray, foam, dry chemical powder and carbon oxide |
| Fire fighting instruction | : | Fire fighters should wear full-faced self contained breathing apparatus and protective clothing |
| Special hazards | : | Not particular |

SECTION 6: ACCIDENTAL RELEASE MEASURES

- | | | |
|-------------------|---|---|
| Leak and/or Spill | : | Wear protective clothing. Eliminate ignition source. Ventilate area. Absorb spills with inert material such as clay, sand, earth, sawdust etc. and collect in a drum. Cover up the contaminated area with household detergent and small amount of water. Brush the slurry and spread inert absorbents on the slurry liquid and collect the absorbed material in a drum. Seal drum and dispose of. Do not contaminate water resources. |
|-------------------|---|---|

SECTION 7: HANDLING AND STORAGE

- | | | |
|----------|---|--|
| Handling | : | Keep out of reach of children. Avoid contact with the eyes, skin and clothing and avoid inhalation of product or spray mist. If in eyes, wash it immediately with water. After handling and before eating, drinking or smoking, wash hands, arms and face thoroughly with soap and water. Do not contaminate food or use on household tanks. |
| Storage | : | Store in the closed, original container in a dry, cool, well-ventilated area, secure area at minimum storage temperatures. Store in locked room or place away from children, animals, food, animal feed, seed and fertilizers. |

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

- | | | |
|----------------------|---|--|
| Exposure Limit | : | No occupational exposure limits established by OSHA, ACGIH or NIOSH. |
| Engineering measures | : | Local exhaust ventilation |
| Personal protection | : | Wear pesticide respiratory masks, protective gloves and clothing. |



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SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

Appearance	: Brown liquid
pH value (1 % aq. sol.)	: 5 – 7
Water (%)	: ≤ 0.5
Emulsion stability (at 2 hr.,ml)	: “Cream ≤ 4 ”, “Oil ≤ 0.5 ”

SECTION 10: STABILITY AND REACTIVITY

Conditions to avoid	: Direct sunlight, heat and extreme temperature
Incompatible	: Carbamate and organophosphorous pesticides.
Decomposition products	: No data available
Hazardous polymerization	: No data available
Stability	: Stable under normal condition.

SECTION 11: TOXICOLOGICAL INFORMATION

CLOMAZONE:

Oral Acute oral LD_{50} for male rats 2077, female rats 1369 mg/kg. **Skin and eye** Acute percutaneous LD_{50} for rabbits >2000 mg/kg. Practically non-irritating to eyes and skin (rabbits). Not a skin sensitiser (guinea pigs). **Inhalation** LC_{50} (4 h) for rats 4.8 mg/l. **NOEL** (2 y) for rats 4.3 mg/kg daily; (1 y) for dogs 13.3–14 mg/kg b.w. daily (*EFSA Sci. Rep.*). **ADI/RfD** (EC) 0.133 mg/kg b.w. [2007]; (BfR) 0.043 mg/kg b.w. [2003]; (EPA) 0.043 mg/kg b.w. [1987]. **Toxicity Class** WHO (a.i.) II; EPA (formulation) III

PROPANIL:

Oral Acute oral LD_{50} for rats >2500, mice c. 1800 mg/kg. **Skin and eye** Acute percutaneous LD_{50} (24 h) for rats >5000 mg/kg. Not irritating to skin and eyes (rabbits). Not a skin sensitiser (guinea pigs). **Inhalation** LC_{50} (4 h) for rats >1.25 mg/l air (dust). **NOEL** (2 y) for rats 400, dogs 600 mg/kg diet. **ADI/RfD** (EPA) RfD 0.009 mg/kg b.w. [2006]. **Water GV** GV not established; readily transformed into metabolites that are more toxic; a GV for the parent compound is considered inappropriate, and there are inadequate data to enable the derivation of GVs for the metabolites. **Other** Not mutagenic, not carcinogenic. **Toxicity Class** WHO (a.i.) III; EPA (formulation) III (EC) **EC classification** Xn; R22| N; R50| concn. dep.

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SECTION 12: ECOLOGICAL INFORMATION

CLOMAZONE:

Birds Acute oral LD_{50} for bobwhite quail and mallard ducks >2510 mg/kg. LC_{50} (8 d) for bobwhite quail and mallard ducks >5620 ppm. **Fish** LC_{50} (96 h) for bluegill sunfish 34, Atlantic silverside 6.26, rainbow trout 19 mg/l. **Daphnia** LC_{50} (48 h) 5.2 mg/l. **Algae** For *Navicula pelliculosa*, E_bC_{50} 0.136 mg/l, E_rC_{50} >0.185 mg/l (*EFSA Sci. Rep.*). For *Selenastrum capricornutum*, E_bC_{50} 2.0 mg/l, E_rC_{50} 4.1 mg/l (*EFSA Sci. Rep.*). **Other aquatic spp.** LC_{50} (96 h) for mysid shrimps (*Mysidopsis bahia*) 0.57, Eastern oysters (*Crassostrea variegatus*) 5.3 mg/l. **Bees** Acute LD_{50} for honeybees (oral) >85.29, (contact) >100.0 µg/bee (*EFSA Sci. Rep.*). **Worms** LC_{50} (14 d) for *Eisenia foetida* 78 mg/kg soil (corrected, because $\log P > 2$) (*EFSA Sci. Rep.*).



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PROPANIL:

Birds Acute oral LD_{50} for mallard ducks 375, bobwhite quail 196 mg/kg. Dietary LC_{50} (5 d) for mallard ducks 5627, bobwhite quail 2861 ppm. **Fish** LC_{50} (48 h) for carp 8–11 mg/l. **Daphnia** LC_{50} (48 h) 4.8 mg/l.

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SECTION 13: DISPOSAL CONSIDERATIONS

Do not reuse product containers. Dispose of product containers, waste containers, and residues according to local, state and environmental regulations.

SECTION 14: TRANSPORTATION INFORMATION

Class No.	: 9
UN No.	: 3082
Packaging group	: III
Dangerous Goods Class	: Class 3, Flammable liquids shall not be loaded in the same vehicle or packed

SECTION 15: REGULATORY INFORMATION

Follow all regulations in your country.

SECTION 16: OTHER INFORMATION

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